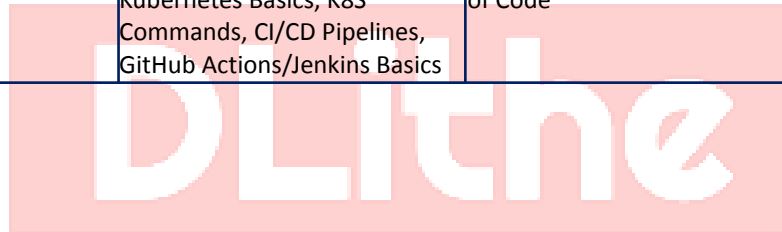


LANGUAGE: JAVA PROGRAMMING LEVEL: BASIC-INTERMEDIATE

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	Corporate-Level Use Case
Day 1	Java Fundamentals	JVM, JDK, JRE, Data Types, Variables, Operators, Input/Output, Type Casting, Control Statements	LeetCode: Palindrome Number, Fizz Buzz HackerRank: Java Stdin and Stdout, Java If-Else	Enterprise Employee Onboarding System
Day 2	Arrays, Strings & Methods	Arrays, 2D Arrays, String Class, StringBuilder, Methods, Method Overloading, Pattern Programs	LeetCode: Two Sum, Maximum Subarray, Valid Anagram HackerRank: Java Arrays, Java Strings Introduction	Customer Feedback Analysis Engine
Day 3	OOP Fundamentals	Classes & Objects, Constructors, this Keyword, Static Keyword, Encapsulation	LeetCode: Climbing Stairs, Merge Sorted Array HackerRank: Java Methods, Java Static Initializer Block	HR Management Portal
Day 4	Advanced OOP Concepts	Inheritance, Polymorphism, Abstraction, Interfaces, Method Overriding, Runtime Polymorphism	LeetCode: Valid Parentheses, Single Number HackerRank: Inheritance, Abstract Classes	Banking Transaction Processing System
Day 5	Exception Handling & Packages	try-catch-finally, throw/throws, Custom Exceptions, Packages, Access Modifiers, Logging Basics	LeetCode: Contains Duplicate, Missing Number HackerRank: Java Exception Handling	Secure Payment Gateway Validation System
Day 6	Collections Framework – Part 1	ArrayList, LinkedList, Stack, Queue, Vector, Iterators	LeetCode: Implement Queue using Stacks, Valid Parentheses HackerRank: Java List, Java Stack	Inventory & Warehouse Tracking System
Day 7	Collections Framework – Part 2	HashMap, TreeMap, HashSet, TreeSet, Comparable vs Comparator	LeetCode: Group Anagrams, Top K Frequent Elements, Majority Element HackerRank: Java Map, Java Hashset	Fraud Detection & Transaction Monitoring System
Day 8	File Handling & Multithreading	File Handling, BufferedReader, FileWriter, Serialization Basics, Threads, Runnable, Synchronization	LeetCode: Print in Order, FizzBuzz Multithreaded HackerRank: Java Threads	Multi-threaded Server Log Processing System
Day 9	Java 8 Features	Lambda Expressions, Functional Interfaces, Stream API, Optional Class, Method References	LeetCode: Merge Intervals, Binary Search HackerRank: Java Lambda Expressions	Real-Time Data Analytics Pipeline
Day 10	JDBC & Database Connectivity	JDBC Architecture, MySQL Connectivity, CRUD Operations, PreparedStatement, Transactions	LeetCode: Rotate Array, Move Zeroes HackerRank: Java JDBC Concepts	Enterprise Employee Database Management System
Day 11	Advanced Java Concepts	Generics, Wrapper Classes, Immutable Class, Singleton Design Pattern, SOLID Principles Basics	LeetCode: LRU Cache, Valid Sudoku HackerRank: Java Generics	Enterprise Configuration Management System
Day 12	Advanced Interview Coding Practice	Time Complexity, Space Complexity, Sliding Window, Hashing, Searching & Sorting,	LeetCode: Longest Substring Without Repeating Characters, Best Time to Buy	Campus Interview Coding Assessment Platform

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	Corporate-Level Use Case
		Debugging, Competitive Coding Patterns	& Sell Stock, Merge Intervals, Binary Search HackerRank: Balanced Brackets, Sparse Arrays, Left Rotation	
Day 13	Advanced Coding + RDBMS Concepts	First Half – Coding Practice: Mock Interview Round, Optimization Techniques, Collections-Based Coding, Debugging Round. Second Half – RDBMS Concepts: DBMS vs RDBMS, ER Model, Keys, Normalization, SQL Basics, SELECT, INSERT, UPDATE, DELETE, Joins, Aggregate Functions	LeetCode: Contains Duplicate, Rotate Array, Valid Anagram HackerRank: SQL Basic Select, Dynamic Array, Weather Observation Station	Enterprise Resource Management Database System
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Cloud Computing Basics, IaaS/PaaS/SaaS, Git & GitHub Workflow, clone/push/pull/merge conflicts, Docker Basics, Docker Images & Containers, Kubernetes Basics, K8S Commands, CI/CD Pipelines, GitHub Actions/Jenkins Basics	LeetCode: Single Number, Move Zeroes, Merge Sorted Array HackerRank: Linux Shell Basics, Git Tutorials, 30 Days of Code	End-to-End Microservices Deployment Pipeline using GitHub + Docker + Kubernetes



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